Kinetic-Nexus: Build Specification

Written for AI assistants helping with the project. Version 1.1. Designed in SolidWorks, printed in PETG on an FDM printer. All lengths are in millimetres.

0. Read this first (instructions to the AI assistant)

You are helping one maker design, model in SolidWorks, 3D print and assemble a project called Kinetic-Nexus. This document is the single source of truth.

- Treat every number in sections 4 to 9 as the current baseline (v1). They are first-pass engineering estimates, not proven values. Test prints (section 13) decide the final values.
- Never change a baseline number silently. If you propose a change, state the old value, the new value, the reason, and everything that moves as a result (for example collapsed and extended length).
- Keep every dimension as a named parameter (section 2.6). Do not hard-code values.
- If an answer depends on an open item (section 15), ask the maker for it before generating geometry. Ask one question at a time.
- Lead the maker one build step at a time (section 12). Name the exact SolidWorks tool or feature. If you are not sure a tool exists in their edition (for example Toolbox), say so and give a fallback.
- Stay inside the $30 hardware budget (section 10) and the print process (section 11). If you add a part, state its cost.
- If this document is unclear or contradicts itself, stop and ask. Do not guess.

What changed in version 1.1 (only relevant if you saw version 1.0):

- The pull-out stop is now a separate stop band placed above the tube's bottom end (section 5.3). Version 1.0 used the bottom guide band for both guiding and stopping, which would have cut the overlap to about 6 and made the length math wrong.
- Stop lip engagement raised from 0.275 to 0.45 per face (`relief` 0.6, `lip_clr` 0.30), with stronger fallbacks in section 14.
- Collet fingers lengthened from 6 (an 8 collet with a 2 lip ring) to 12, because short printed fingers would crack. `collar_h` grew from 16 to 22 and T1 shrank from 160 to 145 to hold the collapsed length. Collapsed overall is now 256 (was 253) and extended overall is 670 (was 685).

- A build-height check is now step 0, and each open item says when it is needed (section 15).
- Tip sag estimate is now 7 to 11 mm and joint play about 14 mm; coupon C1 also tests flexure tabs.

1. What is being built

Kinetic-Nexus is a pocket-sized, 3D-printed telescoping selfie stick and vlogging baton that holds a smartphone. It is a hobby hardware project for Hack Club Half-Life (a hardware engineering program), designed in SolidWorks over roughly 15 to 22 hours with a hardware budget of $30. Every structural part is printed in PETG on a desktop FDM printer.

What it looks like: a grip, a stack of three stepped round collars above it (like a zoom lens barrel), four hexagonal tubes nested inside each other, and a phone-holding head on top.

Features:

- F1 Four-segment hexagonal telescoping pole. The hex shape stops the tubes twisting against each other.
- F2 An independent lock at every joint: a split collet plus a conical threaded nut (3 joints).
- F3 Preset detent clicks at 0, 50 and 100 percent of each joint's travel, plus a ruler scale on one flat of each tube.
- F4 Worm-gear phone tilt with a thumb-wheel. It is self-locking, so the phone cannot sag.
- F5 Self-centering twin-rack phone clamp with a printed flexure ratchet and silicone pads. This is the hero mechanical movement: one pinch moves both jaws equally, so the phone is always centered on the stick.
- F6 Three tripod legs that fold flat against the grip and flip out.
- F7 A hidden bay in the grip for an off-the-shelf Bluetooth shutter remote, with a thumb button.
- F8 Accessory mounts: a 1/4-20 tripod socket in the base, plus a 1/4-20 insert and a cold shoe on the head.

2. Definitions and conventions (read carefully)

- 2.1 Units are millimetres and degrees.

- 2.2 AF means across flats: the distance between two parallel flat faces of a hexagon. Corner-to-corner (CC) = AF × 1.1547. 'Hex size' always means AF.
- 2.3 Clearance `clr` is diametral across flats: the AF of the bore minus the AF of the part sliding in it. The gap per face is `clr` /2. Baseline `clr` = 0.45 (0.225 per face).
- 2.4 Names. T1 is the outermost tube (the grip tube). T2 and T3 are the middle tubes. T4 is the innermost (tip) tube. J1 is the joint between T1 and T2, J2 between T2 and T3, J3 between T3 and T4. A collar is a collet plus nut at the top of a tube: collar 1 on T1, collar 2 on T2, collar 3 on T3. T4 has no collar.
- 2.5 Axis. z runs along the stick. z = 0 is the bottom face of the grip body. Positive z points toward the phone. A foot pad extends below z = 0 down to z = -8 (section 8.1).
- 2.6 Parameters (name = baseline): `AF1` = 32, `wall` = 2.0, `clr` = 0.45, `step` = 2 × `wall` + `clr` = 4.45, `tail_h` = 6, `stop_h` = 6, `relief` = 0.6 per face, `lip_clr` = 0.30, `floor_t` = 2, `boss_h` = 8, `finger_L` = 12, `collet_h` = 14, `collar_h` = 22, `thread_pitch` = 3.0, `thread_depth` = 1.4, `cone_half_angle` = 8, `head_h` = 45, `foot_h` = 8, `leg_len` = 90, `leg_angle` = 40, `p_leg` = 22, `jaw_min` = 58, `jaw_max` = 92.
- 2.7 Pole length is z from 0 to the top of the head. Overall length is pole length plus `foot_h`.
- 2.8 Overlap `o` of an inner tube: when fully extended, the distance from the mid-height of the collet above it to the bottom end of that inner tube. Stroke: how far an inner tube moves relative to the tube outside it between collapsed and extended. Guide band, stop band and body are defined in section 5.3.
- 2.9 Stowed pose: pole collapsed, legs folded, clamp closed, head pointing along the stick.

3. Requirements

Hard requirements:

- R1 Collapsed overall length at most 300. Target at most 260.
- R2 Extended overall length as long as possible. Target at least 650.
- R3 Hardware cost at most $30 (filament, printer and tools are not counted).
- R4 FDM PETG, 0.4 mm nozzle, minimal supports.
- R5 Fully parametric in SolidWorks.
- R6 Every telescoping joint locks independently.

- R7 Holds a 250 g phone without slipping or visible droop. Phone width 58 to 92, thickness up to 14 (case on).
- R8 Pocketable. The grip is about 44 across corners; the maker accepts this.

Out of scope: electronics other than the off-the-shelf remote, waterproofing, any metal parts other than screws, inserts and magnets.

4. Architecture and length math

Stowed, bottom to top: foot pad (z -8 to 0), grip body (z 0 to 110, AF 38), T1 neck (z 110 to 145, AF 32), collar 1 (z 137 to 159, OD 44), collar 2 (z 159 to 181, OD 39), collar 3 (z 181 to 203, OD 34), head (45 tall, z 203 to 248).

Why the collars stack: each collar is wider than the tube outside it, so an inner tube cannot slide fully inside. Its collar rests on top of the previous collar. Each joint therefore adds `collar_h` = 22 to the collapsed length (8 of thread boss plus 14 of collet).

Results:

- Collapsed: pole 248, overall 256.
- Extended: pole 662, overall 670.
- Preset clicks (all joints at the same fraction): pole 248 / 455 / 662, overall 256 / 463 / 670.
- Strokes: J1 = 110, J2 = 138, J3 = 166 (sum 414). Extended pole = collapsed pole + stroke sum.

Chain check of the extended length (z in mm): collet 1 mid-height = 145 + 7 = 152. T2 bottom = 152 - 40 = 112, T2 top = 112 + 165 = 277. Collet 2 mid = 277 + 7 = 284. T3 bottom = 284 - 34 = 250, T3 top = 250 + 187 = 437. Collet 3 mid = 437 + 7 = 444. T4 bottom = 444 - 28 = 416, T4 top = 416 + 201 = 617. Head top = 617 + 45 = 662.

If `head_h`, `foot_h`, `collar_h` or any tube length changes, recompute section 5.2 first, then these results.

5. Tube nest

5.1 Sizing rule: AF(n+1) = AF(n) - `step`. Bore AF = AF - 2 × `wall`.

5.2 Tube data. AF is the full size (the band size for inner tubes). z values are measured from the grip body bottom.

- T1: AF 32.00 (CC 36.95). Bore 28.00 (CC 32.33). Length 145, z 0 to 145. A 2 mm floor sits at z 0 to 2.
- T2: AF 27.55 (CC 31.81). Bore 23.55 (CC 27.19). Length 165. Collapsed z 2 to 167. Overlap 40. Extended z 112 to 277. Stroke 110.
- T3: AF 23.10 (CC 26.67). Bore 19.10 (CC 22.05). Length 187. Collapsed z 2 to 189. Overlap 34. Extended z 250 to 437. Stroke 138.
- T4: AF 18.65 (CC 21.54). Bore 14.65 (CC 16.92). Length 201. Collapsed z 2 to 203. Overlap 28. Extended z 416 to 617. Stroke 166.

Overlap is about 1.5 × the tube's AF on purpose: it limits wobble.

5.3 Inner-tube construction (T2, T3, T4):

- Guide band: the bottom `tail_h` (6) of the tube, at full AF. It slides in the bore of the tube outside it (clearance `clr`).
- Stop band: a second full-AF band, `stop_h` (6) tall. Its top face sits D above the tube's bottom end, where D = overlap - 7 (T2 = 33, T3 = 27, T4 = 21); the 7 is the distance from a tube's top end to the collet's mid-height. When the tube is pulled out, this top face hits the underside of the collet's lip ring. This band, not the guide band, is the pull-out stop, and it is what sets the overlap.
- Body: everything not in a band, relieved by `relief` per face (body AF = AF - 1.2). Net body wall = 2.0 - 0.6 = 1.4. The body passes through the collet bore.
- Lip: the collet's solid 2 mm lip ring has a hex bore of body AF + `lip_clr` = AF - 0.9, so the stop band overlaps it by 0.45 per face. Printer error of 0.10 to 0.15 per feature can still remove a third to a half of that, so prove the stop in coupon C2 before relying on it (fallbacks in section 14).
- Crush ribs: on each guide-band face, two vertical ribs 0.5 wide and 0.15 proud, to preload the fit and remove wobble. Fixed ribs are sensitive to printer calibration, so coupon C1 also tests printed flexure tabs; use whichever is more consistent.
- Chamfer 0.8 on every tube mouth and bottom end (hides elephant's foot). Round the outer hex corners of both bands with 0.5 radius so FDM-rounded bore corners do not bind.
- Top ends of T1, T2, T3 carry a round thread boss (section 6). The top end of T4 takes the head stem (section 9.4).

5.4 Stiffness estimate (first-pass hand calculation, not simulation). Hex tube second moment: I = (5 × √3 / 16) × (s_out^4 - s_in^4) with s = AF / √3, using the relieved body AF. Load 2.45 N (a 250 g phone) at about z = 650, PETG modulus 1.2 to 1.8 GPa, joints rigid.

Result: tip deflection about 7 to 11 mm, peak bending stress about 1.7 MPa (far below PETG layer strength of about 25 MPa). The thin T4 is the weak link.

5.5 Joint play. When a collet clamps the mouth, the guide band can still shift by `clr`. Angle = `clr` / overlap. Tip movement = angle × distance from joint to phone. With no preload: J1 about 5.6, J2 about 4.8, J3 about 3.3, total about 14 mm worst case. The crush ribs (or flexure tabs) exist to remove this.

6. Collets and nuts (J1, J2, J3)

Each joint is three parts: a thread boss on the top of the outer tube, a separate collet that sits on that tube's top end face, and a nut.

- Thread boss: round, top `boss_h` = 8 mm of the tube (it blends from the hex with a 45 degree transition). Major diameters: J1 38.1, J2 33.0, J3 27.9. Root diameters: 35.3, 30.2, 25.05. Reason for round: a hex corner at the bore would leave only 0.65 wall under a smaller thread.
- Thread: pitch 3.0, depth 1.4, trapezoidal, 20 degree flanks, 1.0 flat on crest and root, radial clearance 0.3 per side, 2.5 turns (8 mm) engaged.
- Collet: `collet_h` = 14 tall above the tube end. A solid 2 mm lip ring at the bottom, then six fingers (`finger_L` = 12), one per hex flat, 2.0 thick. Slit the fingers at the hex corners, 0.8 wide, from the top of the collet down to the lip ring only; the lip ring stays unslit. The outer surface is a cone with `cone_half_angle` 8, wider at the base. The bore (lip ring and fingers) = body AF + `lip_clr` (0.30), which equals tube AF - 0.9.
- Nut: height `collar_h` = 22 (8 thread plus 14 cone). Internal cone matches the collet cone. Outer diameters: J1 44, J2 39, J3 34. Add 24 shallow grip flutes.
- Lock action: the free gap is `lip_clr` / 2 = 0.15 per face, and the fingers must close about 0.2. Axial nut travel = 0.2 / tan(8 degrees) = 1.4, which is about 0.5 turn of a 3 mm thread. Each joint locks in about half a turn.
- Finger stress and hold (estimates): a finger 12 long and 2.0 thick deflected 0.2 sees about 6 MPa (3 × E × t × deflection / (2 × L squared), E about 1.5 GPa). Fingers of 6 to 8 long would see 14 to 25 MPa and crack, so do not shorten `finger_L`. Six fingers give about 25 N of clamp force and roughly 7 N of axial hold, rising toward 15 N with extra nut turn. The phone plus tubes weigh about 4 N, so this is enough, but prove it in coupon C2.
- A hard stop on the nut prevents over-tightening that would crack the fingers.

- Stack rule: when collapsed, the thread boss of T2 sits on the top face of collar 1 and the boss of T3 on collar 2. T4′s top end finishes flush with the top of collar 3.

7. Detents and ruler

- Each collet has a printed flexure tooth: a cantilever 12 long, 3 wide, 1.2 thick, with a bump of 0.8 radius proud by 0.5. Estimated click force about 0.4 N.
- Each inner tube has three V dimples (60 degree, 0.4 deep) on flat 1 of its body, at stroke fractions 0, 0.5 and 1.0.
- Dimple distance below the tube′s top end: T2 = 15, 70, 125. T3 = 15, 84, 153. T4 = 7, 90, 173. (Formula: offset + fraction × stroke, where offset is the tube′s top-end height minus the collet′s mid-height when collapsed: 15 for T2 and T3, 7 for T4.)
- Ruler: embossed 0.2 proud (inside the relief zone) on flat 1. A tick every 10, a long tick every 50, numbers every 50 in 3 mm text. Zero tick is where the collar′s top face touches the tube when collapsed: 8 below the top end for T2 and T3, 0 for T4. The tick level with the collar top face is the extension in mm.

8. Grip, base, legs, remote bay

8.1 Grip body and foot pad. One printed part with T1. Outer is AF 38 (CC 43.9) from z 0 to 110, then AF 32 above. Bore AF 28 throughout. Foot pad: a round pad of about 20 diameter from z -8 to 0 carrying the 1/4-20 tripod insert. The insert must be at most 9 long so it ends below the floor top (z 2). Insert hole size comes from the insert vendor′s datasheet, not from guesswork. The pad adds `foot_h` = 8 to overall length (already counted in section 4).

8.2 Legs (3). Leg 90 × 14 × 4. They lie in pockets 2.5 deep on alternating flats (flats 1, 3, 5) of the grip, stowed pointing up (+z). Hinge: M3 screw with nyloc nut, pivot axis at z = 10 and distance `p_leg` = 22 from the stick axis (hinge ears at the grip bottom). Deploying rotates each leg 140 degrees outward and down to a hard stop at `leg_angle` = 40 degrees from the downward vertical.

- Foot position: 90 × cos(40) = 69 below the pivot, so feet are about 59 below z = 0 (about 51 below the foot pad). Foot radius R = 22 + 90 × sin(40) = 80. Footprint is a triangle with circumradius 80, inradius 40.
- Stability: tips over at a lean of about atan(40 / CG height), roughly 4 to 5 degrees at the 455 preset with a 250 g phone. Free-standing on a desk is acceptable up to the 455 preset. At full length it is hand-held only.

- Stow latch: 2 magnets (3 × 2) per leg, one in the leg and one in the grip pocket, polarity checked. Keep a 0.4 to 0.6 cap layer over each magnet.

8.3 Remote bay. A thumb blister on flat 2 of the grip, about 12 proud. Drawer pocket (placeholder, confirm with the real remote): 46 × 26 × 11 plus 0.3 clearance. The drawer slides in from the bottom face and is held by 2 magnets. A printed flexure button (cantilever pad about 12 diameter, 0.8 travel) presses the remote's own button. The remote slides out to change its coin cell.

9. Head

9.1 Size. `head_h` = 45 along the stick in the stowed pose. Width about 74 with the clamp closed and about 108 fully open; depth about 25. Confirm in CAD.

9.2 Worm tilt drive.

- Module 1.25, 20 degree pressure angle. Single-start worm, pitch diameter 10, about 14 long. Wheel is a sector: 24-tooth wheel (pitch diameter 30) cut to a 135 degree arc (about 10 teeth), face width 8. Center distance 20.
- Ratio 24:1, so 15 degrees of tilt per thumb-wheel turn (about 9 turns end to end). Hard stops at both ends of the sector.
- Lead angle = atan(lead / (π × 10)) with lead = π × 1.25 = 3.93, giving 7.1 degrees. It self-locks when friction coefficient exceeds tan(7.1) = 0.125. Printed PETG on PETG is roughly 0.25 to 0.4. Do not add grease: it can destroy self-locking.
- Load check: 250 g phone, 40 arm = 98 N·mm, tooth force about 6.5 N, Lewis stress about 2 MPa.
- Thumb-wheel: 24 diameter with grip flutes, printed as one piece with the worm shaft. The shaft sits in two printed bosses with thrust faces on both sides. Backlash target 0.15.
- Tilt axis: one M3 × 20 stainless screw with nyloc nut. Sector bore 3.2.

9.3 Clamp (twin rack, self-centering).

- Gap opens from `jaw_min` 58 to `jaw_max` 92, so 17 travel per jaw.
- Two racks (module 1.5, 20 degree pressure angle, about 30 long) mesh a 14-tooth pinion (pitch diameter 21) on a fixed axle. Pinion turns 17 / 10.5 rad = about 93 degrees across the full range. Both jaws move equally, so the phone stays centered.
- Each jaw is a rack plus an L-shaped lip. Phone thickness capacity 14.

- Ratchet: teeth of 1.5 pitch on the back of one rack, engaged by a printed flexure pawl (1.2 thick × 14 long, estimated 0.85 N). Each click closes the gap by 2 × 1.5 = 3. A release tab lifts the pawl.
- Pads: 4 mm silicone sheet on each jaw face to absorb the 3 mm step and add grip.
- Reason for a ratchet: a printed PETG spring tops out near 2 N, but a phone needs 8 N or more. The ratchet lets the user's pinch set the force.
- Fallback if the ratchet misbehaves: a rubber band or small extension spring between the jaws (about $2).
- Include a viewing window of about 12 × 10 over the pinion so the gear is visible.

9.4 Mounting and accessories. Proposed (confirm in CAD): hex stem AF 14.25, 16 long, press-fit into the T4 bore and held by an M3 × 12 cross screw into a nut trap. The head top carries a 1/4-20 brass insert and a printed cold shoe. Copy the cold shoe profile from the ISO 518 standard or a measured commercial shoe; do not guess its dimensions.

10. Bill of materials and budget

- 1/4-20 brass heat-set inserts, pack of 10. Uses 2 (base and head). About $6.
- M3 screw and nyloc nut assortment. Uses 3 × M3 × 16 (leg hinges), 1 × M3 × 20 (tilt axis), 1 × M3 × 12 (head stem), 5 nuts. About $7.
- Neodymium disc magnets 3 × 2, pack of 12 to 20. Uses 8 (6 for legs, 2 for the remote drawer). About $5.
- Bluetooth shutter remote, 1. About $5.
- Silicone or rubber sheet 4 mm thick for the jaw pads. About $3.
- Estimated total about $26 of $30, leaving about $4 reserve.

Prices are rough US estimates. The maker must check current prices, shipping, tax, and what Hack Club Half-Life counts toward the $30. Deliberately not purchased: springs, steel balls, bearings, rods. Detents, flexures and pivots are printed.

11. Printing and DFM rules

Typical PETG starting values (verify against the filament datasheet): nozzle 0.4, layer 0.2 (0.12 for the worm, 0.16 for threads), nozzle 230 to 245 C, bed 70 to 80 C, part cooling 30 to 50 percent. Tubes use 4 to 5 perimeters and no infill. Small solid parts (gears, collets, legs) at 100 percent infill. Seam placed on a hex corner. No supports by design.

Orientation:

- T1 with grip, T2, T3, T4: vertical (axis along Z), tail end down, for best hex accuracy.
- Nuts: vertical (thread axis along Z).
- Collets: on their side with a hex flat facing up (so the bore roof is a short bridge, not a steep overhang), so finger bending stress runs along the filament lines, not across layers. Collets carry no threads, only a hex bore and an outer cone, so side printing distorts little. If coupon C2 shows a warped cone or bore, print the collet vertically instead (the 12 long fingers then see about 6 MPa across layers) and compare.
- Worm: vertical, 0.12 layers. Sector, racks, jaws, housing halves and legs: flat.
- Why: layer lines are the weak direction. Do not put a high tensile or bending load across layers.

Build height preflight (step 0, before any CAD): T4 is 201 tall and T3 is 187. The printer needs at least 210 of usable Z, meaning the listed build height minus any gantry or nozzle clearance (a brim adds only a layer or two). Tall thin tubes wobble, so print T4 with a brim and a slower outer wall. If the printer has less height, ask the maker before changing the design. The planned fallback is to split T3 and T4 into two sections each, joined by a 30 long hex splice sleeve, printed upright and glued.

PETG softens near 70 to 80 C, which survives a hot car better than PLA (55 to 60 C). PETG also creeps under sustained clamp load, so do not store the stick with collars cranked tight.

About 25 printed parts in total.

12. SolidWorks build plan (about 20 hours)

Lead the maker through these in order, one at a time.

Step 0 (about 15 minutes, before step 1): confirm the printer's usable build height is at least 210, and which SolidWorks edition and add-ins the maker has. The other open items in section 15 are needed later, each at the step named there.

1. Equations and tube part (2 h). Tools > Equations: create the parameters in section 2.6 plus `AFn = AF1 - (n-1) × step`. Sketch a 6-sided Polygon using the inscribed-circle option (circle touching the flats) and dimension its diameter to AF. Verify flat-to-flat with Measure. Extrude to length, offset inward by `wall`, cut. Insert > Tables > Design Table to make T1 to T4 as configurations. Then print a 2-joint coupon (section 13) before going further.
2. Collets, nuts, thread bosses (3 h). Use Insert > Features > Thread or a swept cut for the trapezoid profile. Make J1 to J3 configurations.

3. Grip body, legs, foot pad, remote bay (4 h). Confirm the remote's size first (section 15).
4. Head, worm and sector (4 h). If Toolbox is not available, model teeth with an equation-driven curve or import a gear from a generator and check the mesh.
5. Clamp (3 h). Rack and pinion, pawl, jaws, window.
6. Detents and ruler (1 h).
7. Assembly, motion study and checks (3 h). Mate hex flats with parallel and distance mates and use Limit Distance mates to model travel. Make Collapsed and Extended configurations. Run Tools > Evaluate > Interference Detection. Run a Motion Study for the worm, clamp and legs. Make an exploded view and renders (PhotoView 360 if available).

Cut order if time runs short: ruler first, then remote bay, then leg magnets, then detents. Never cut the collets or the tube nest.

13. Validation: test prints and acceptance

- C1 Clearance coupon: four short pairs of nested hex sleeves at `clr` = 0.35, 0.45, 0.55, 0.65, each with crush ribs at 0.10, 0.20 and 0.30 proud, a printed flexure-tab variant (leaf spring 0.8 thick, 10 long, 0.3 proud) and a no-rib control. Pick the loosest value that has no wobble and no binding.
- C2 Thread and collet coupon: a 60 mm section of T1 and T2 with collet and nut. Pass: locks in about half a turn (0.4 to 0.8), releases easily, no binding, holds a 15 N axial load without slipping, the stop band holds a 50 N pull without the lip chipping, and the fingers show no cracks after 50 lock and unlock cycles.
- C3 Gear coupon: worm, sector and housing. Pass: backlash 0.1 to 0.2, and a 250 g weight on a 40 arm holds for 60 seconds with no drift.
- C4 Clamp coupon: racks, pinion, pawl. Pass: clicks cleanly, holds a 300 g dummy phone for 10 minutes.
- Final acceptance: collapsed overall at most 260, extended overall 670 plus or minus 5, tip deflection with 250 g at most 15 including play, no joint creeps over 10 minutes under load, detents click, stands on legs at the 455 preset with a 250 g dummy, fits a jacket pocket.

14. Risks and fallbacks

- Joint wobble about 14 worst case: crush ribs or flexure tabs (coupon C1), retune `clr`, last resort stick-on TPU pads.

- Tip sag 7 to 11: raise `wall` to 2.4 (recompute `step`), raise `AF1` to 34, shorten T4, or accept.
- Stop lip engagement is 0.45 per face and printer error can erode it. If C2 shows chipping or slipping: raise `relief` to 0.75 per face with `wall` 2.4 and `AF1` 34 (engagement 0.6 per face), or add a printed stop ring, or add a corner key (a lug on the stop band running in a groove cut into one bore corner, closed at the top by a nub on the collet).
- Collet fingers crack or hold too little: keep `finger_L` at 12 or more (fingers of 6 to 8 would see 14 to 25 MPa). Each extra mm of `finger_L` adds 3 mm to the collapsed length. If hold stays under 15 N, thicken the fingers to 2.4 and retest.
- Collet printed on its side warps: print it vertically instead and compare in C2.
- Threads bind or strip: keep the coarse thread, test C2, keep the nut hard stop.
- Creep at locked collets and detents: advise not storing collars tight.
- Clamp force too low: thicker pads or finer ratchet pitch (1.0), or the spring fallback.
- Magnets weak through the cap layer: keep the cap at 0.4 to 0.6 and rely on the hard stop when deployed.
- Remote does not fit: resize the bay from the real part.
- Printer too short for T4 (201 tall): check in step 0; the fallback is spliced tubes (section 11).
- Budget overrun: drop the remote, then magnets.

15. Open items to confirm with the maker

1. Printer model, usable build height (at least 210 is needed for T4), nozzle size and PETG brand. Needed in step 0.
2. SolidWorks edition, and whether Toolbox and PhotoView are available. Needed in step 0.
3. Bluetooth remote model and exact dimensions (the bay size is a placeholder). Needed before step 3.
4. The heat-set insert vendor's datasheet (hole size and length). Needed before step 3.
5. Largest phone to support (baseline 58 to 92 wide, 14 thick). Needed before step 5.
6. Hack Club Half-Life rules: what counts toward the $30, and what deliverables are required. Needed before buying parts.
7. Whether to attempt the stretch goal below. Decide after step 5.

16. Stretch goal (only if ahead of schedule)

A 6-blade printed iris on a flared collar 1 (about 52 OD) as a lock-status indicator. It must never carry load, and the stick must work fully without it. Skip it if any baseline requirement is at risk.

17. Glossary

- Collet: a split sleeve whose fingers squeeze inward when a conical nut pushes on them.
- Detent: a small bump that clicks into a dimple to mark a position.
- Flexure: a thin printed bending part used as a spring or latch.
- Heat-set insert: a brass threaded insert pressed in with a hot tip.
- FDM: filament 3D printing. DFM: design for manufacturing.
- Rack and pinion: a straight toothed bar driven by a small gear.
- Worm gear: a screw-shaped gear driving a toothed wheel, usually self-locking at low lead angles.